DEPI Digital Egypt Pioneers Initiative



EDUVANCE AI / PROJECT DOCUMENTATION

EDUVANCE AI

Requirements Gathering

Stakeholders, bounded features, use cases and traceability

TRACK	AI & Data Science - Agentic AI and Generative AI System Developer
DOCUMENT	03 - Project Documentation
SOURCE BASIS	Approved Project Plan, existing WBS documents and repository code

01

Requirements baseline

Collected from Plan.md, WBS 2.1 and WBS 2.2

Eduvance AI is intended to transform bounded training documents into a structured, personalized learning experience. Requirements are compiled from existing approved project scope and team design artifacts; this file does not establish new functional scope.

Input bound

Text-based PDF; initial limit of 3 files and 50 pages per course.

Course target

Approximately 3 modules and 6 lessons; editable lesson objectives.

Learning path

Video, course-specific RAG tutor, assessments, learner progress and project mentor.

Implementation boundary

UI screens are designed in WBS 2.2; no implemented frontend is currently visible on main.

Requirements status

A target feature in the project plan is not the same as a delivered feature. The statuses elsewhere in this document reflect the repository snapshot reviewed during drafting.

02

Stakeholders and personas

Two primary journeys specified in WBS 2.2

Actor / stakeholder	Needs explicitly recorded	Primary interactions
Learner (persona: Sami)	Understand dense source content; receive short lessons and referenced answers; self-assess progress.	View lessons, ask tutor, inspect citations, take quizzes, review progress, receive project brief.
Course creator / instructor (persona: Dr. Laila)	Convert supplied handouts into structured learning material; inspect and edit generated objectives.	Upload PDF documents, review generated course structure, edit objectives and approve lesson creation.
Delivery team	Integrate the scoped modules and verify quality, traceability and reliability.	Architecture, AI evaluation and integration based on the WBS and Plan.

The personas in the source WBS use university examples as user-context scenarios. That does not make this DEPI capstone a university graduation project.

03

Functional requirements

Source-derived identifiers for readability only

ID	Feature	Requirement	Source
FR-01	Document ingestion	Accept text-based PDFs within the initial 3-file / 50-page course limit.	Plan §2, WBS 2.1

ID	Feature	Requirement	Source
FR-02	Content understanding	Extract text, chunks, concepts and source anchors from accepted documents.	Plan §§1-3, WBS 3.2
FR-03	Course generation	Build an editable course outline with modules, lessons, prerequisites and objectives.	Plan §2, WBS 3.3
FR-04	Review and approval	Allow creator/instructor review and editing before lesson-asset generation.	Plan §3, WBS 2.2
FR-05	Educational video	Generate short slide-based lesson video with narration and captions.	Plan §2, WBS 2.3
FR-06	Grounded tutoring	Provide course-specific RAG Q&A with source references and appropriate abstention.	Plan §§2, 8; WBS 2.1
FR-07	Assessments	Generate MCQ / true-false items, objective scoring and practical rubrics.	Plan §2, WBS 3.5
FR-08	Progress	Record completion and quiz events; update progress and concept indicators.	Plan §§2-3, WBS 4.1
FR-09	Revision suggestions	Use transparent rules to suggest concepts/lessons for review.	Plan §2
FR-10	Practical project	Suggest bounded project brief, steps and rubric aligned with studied topics.	Plan §2
FR-11	Account isolation	Support basic accounts and ensure users can access only their own courses and documents.	Plan §2, Backend/auth.py
FR-12	Workflow control	Coordinate states, jobs and saved outputs with bounded retries.	Plan §3, WBS 2.1

04

Non-functional requirements

Constraints explicitly in plan and engineering WBS

Concern	Specified requirement	Primary reference
Input limits	Text-based PDF; initial 3-file and 50-page boundary.	Plan §2
Traceability	Link material and tutor support back to the source; WBS specifies document/page/chunk anchors.	Plan §1, WBS 2.1
Accuracy and grounding	Check source-supported answers and abstention using controlled evaluation.	Plan §8
Deterministic results	Use rules for objective grading, progress and revision triggers; do not infer mastery from video viewing.	Plan §§2-3
Resilience	Monitor tasks and retry within the stated maximum of two retries in the controller design.	WBS 2.1

Concern	Specified requirement	Primary reference
Security / isolation	Protected accounts, scoped data access and permitted training materials.	Plan §§2,9
Feasibility	Work within free-only AI-service/tooling budget and measure operational cost.	WBS proposal, Plan §8
Usability	Evaluate core-user task completion with actual users; do not equate usability with learning improvement.	Plan §8
Portability	Reproducible dependency and database setup; PostgreSQL remains a target to validate.	Backend README, Database README

05

User stories and acceptance intent

Rephrased only from the existing use cases

Story	User need
US-01	As a course creator, I want to upload bounded PDF materials so they can be converted into a structured course.
US-02	As a course creator, I want to review and adjust the generated outline before producing lesson assets.
US-03	As a learner, I want to view narrated lessons with captions to study uploaded material.
US-04	As a learner, I want to ask document-grounded questions and inspect source references.
US-05	As a learner, I want the tutor to abstain or clarify when supplied material does not answer my question.
US-06	As a learner, I want objective quizzes and clear feedback to assess lesson understanding.
US-07	As a learner, I want progress indicators and rule-explained revision suggestions.
US-08	As a learner, I want a bounded practical-project brief linked to completed topics.

Acceptance evidence

The formal targets in Plan §8 cover tutor correctness, abstention, assessment review, workflow reliability, usability and operating measurements. They remain planned thresholds until test results are produced.

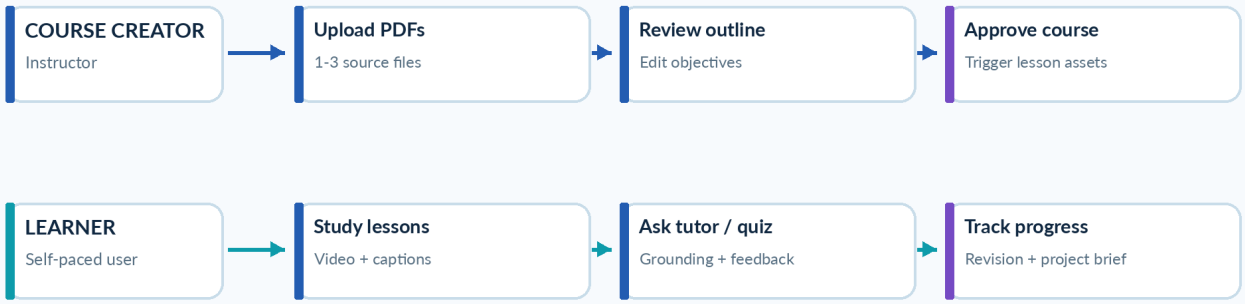
06

User journey and screen mapping

From WBS 2.2 - design, not a released UI

USER JOURNEYS / USE CASE VIEW

Two source-defined actors; screens and actions come from WBS 2.2.



Creator path: prepare and approve. Learner path: study, self-assess, and review.

Actor-to-use-case summary based on the personas and six planned screens in WBS 2.2.

1. Upload & validate
2. Inspect course outline
3. Approve course
4. Study video & tutor
5. Take assessment
6. Review progress & project

Designed screen	Route / interaction	Status of evidence
Upload and ingestion	/upload - document selection and processing progress	Wireframe in WBS 2.2; frontend not present as code
Course review	/course/:id/review - edit lessons and objectives	Wireframe in WBS 2.2; backend course/ workflow routes exist
Lesson player	/course/:id/lesson/:lessonId - video and captions	Wireframe in WBS 2.2; video pipeline code exists
Tutor and citation drawer	Grounded chat with inspected source references	Wireframe in WBS 2.2; Tutor_RAG still placeholder
Quiz and assessment	Lesson quiz and scoring feedback	Wireframe in WBS 2.2; Assessment pipeline code exists
Progress and project	/course/:id/dashboard - progress and project brief	Wireframe in WBS 2.2; backend progress route exists

07

Priorities, exclusions and assumptions

The baseline scope has not been expanded

Priority	Existing Plan.md scope
P0 - Foundation	Upload, extraction, generated course structure, persistence and basic interface.
P1 - Learning experience	Tutor, short videos, assessment, progress and bounded project mentor.
P2 - Optional	Additional languages, richer visual output, broader document support and advanced personalization.

Deferred features: scanned-PDF OCR; broad file formats; advanced knowledge graphs; avatars and advanced animation; unrestricted web research; voice chat; automatic execution/grading of programming submissions; predictive learner models; reinforcement learning; autonomous project supervision; enterprise-scale deployment.

> **Scope constraint**
>
> The plan defines a bounded prototype. Neither the guide nor this report changes the product baseline, member WBS ownership or target thresholds.

08 Requirements traceability

Mapping requirements to existing source artifacts

Requirement group	Specification reference	Available engineering evidence
Accounts / owned courses	FR-11	Backend/auth.py; Backend/courses.py; Backend/tests/
Input and document storage	FR-01, FR-02	Backend/documents.py; AI_Modules/Extraction/
Course design	FR-03, FR-04	AI_Modules/CourseDesign/; WBS 2.2 design
Video lesson generation	FR-05	AI_Modules/Video_Pipeline/ and tests
Tutor with citations	FR-06	WBS 2.1 contract, WBS 2.2 wireframe; Tutor_RAG placeholder
Assessments	FR-07	AI_Modules/Assessment/ and tests
Progress and rules	FR-08, FR-09	Backend/progress.py; Backend/events.py; services/progress_engine.py
Project mentor	FR-10	Plan and WBS contract; standalone module implementation not confirmed
Workflow state control	FR-12	Backend/services/workflow_controller.py; tests

09 Source register and gaps

Material reviewed to produce this requirements draft

S1 - Problem, objectives, scope, priorities and evaluation targets
Plan.md - §§1-3, 8 and 9

S2 - Architecture constraints, traceability and agent contracts
members/member_1/Task_WBS_2.1_Architecture_and_Contracts.md

S3 - Personas, user journey, wireframes and UI tokens
members/member_4/Task_WBS_2.2_UI_UX_Wireframes_and_Journeys.md

S4 - Current implementation evidence
Backend/, Database/, AI_Modules/ on original repository main branch

- A signed stakeholder requirements approval or official graded rubric was not among the supplied materials.
- No new performance latency limit or uptime percentage has been invented.

- Unimplemented screens and RAG components are explicitly marked as designed/targeted rather than delivered.